

WORKSHEET - NAPLAN Year 9

Number and Algebra Solving Equations (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)

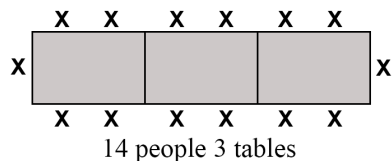
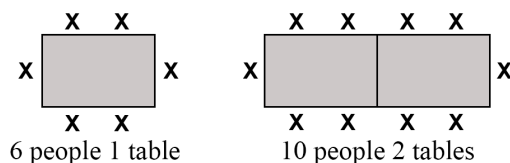


Questions

1. Number and Algebra, NAP-14514

Billy is setting up tables for a comedy night at his club.

An **X** is placed for every available seat at a table, as shown below.



Which of these rules can be used to work out how many people can sit on any row of tables?

- ☐ number of tables $\times 6$
- ☐ number of tables $\div 2 - 2$
- ☐ number of tables $\times 4 + 2$
- ☐ number of tables $\times 4 - 2$

2. Number and Algebra, NAP-08600

In a hockey club, there are 123 players.

There are 21 less women than men.

How many men are in the club?

- ☐ 51
- ☐ 72
- ☐ 81
- ☐ 102

3. Number and Algebra, NAP-32123

David has a job selling mobile phone plans. His weekly salary, W dollars, is calculated using the rule below:

$$W = 300 + 0.05P$$

where P is the total value in dollars of the mobile phone plans he sells that week.

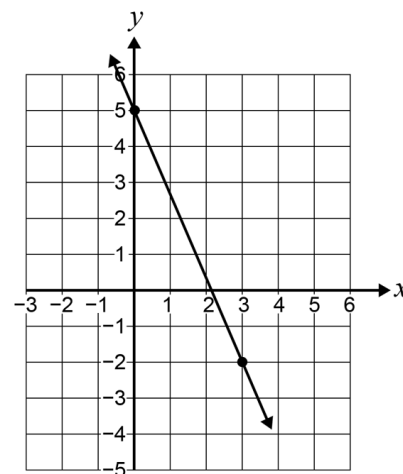
David sold \$32 000 worth of mobile phone plans in a given week.

What was David's salary in the week?

- ☐ \$640
- ☐ \$1280
- ☐ \$1900
- ☐ \$3230
- ☐ \$16 000

4. Number and Algebra, NAP-02715

Leo drew a straight line through the points (0, 5) and (3, -2) as shown in the diagram below.



What is the gradient of the line that Leo drew?

- ☐ $\frac{7}{3}$
- ☐ $\frac{3}{7}$
- ☐ $-\frac{3}{7}$
- ☐ $-\frac{7}{3}$

5. Number and Algebra, NAP-02691

Pamela looks at the price of four sunscreens.

			
Sun sunscreen \$10.90, 1200 mL	Sport sunscreen \$5.40, 600 mL	Block sunscreen \$4.75, 500 mL	Wet sunscreen \$7.40, 750 mL

Which sunscreen is the cheapest per litre?

Sun	Sport	Block	Wet
☐	☐	☐	☐

6. Number and Algebra, NAP-08629

Package Delivery Pty Ltd charges the following for delivering a package with mass x kg

$$\text{Price} = 9 + 4.5x$$

What is the mass, in kg, of a package that costs \$90 to deliver?

10	18	20	22
☐	☐	☐	☐

7. Number and Algebra, NAP-55673

It takes 15 minutes for all of the sand to flow through a sand timer.

The sand timer contains 475 grams of sand.

Each millilitre of sand weighs 1.5 grams.



How many millilitres of sand flow through the timer per minute?

10.3	21.1	31.7	47.5
☐	☐	☐	☐

8. Number and Algebra, NAP-20326

Mr. Soros put \$500 into a simple interest account for a year.

He did not take any money out or add any money to the account.

At the end of the year he had \$530 in the account.

What was the annual percentage interest rate?

 %

Worked Solutions

1. Number and Algebra, NAP-14514

Consider Option 3:

1st table: $1 \times 4 + 2 = 6$ people

2nd table: $2 \times 4 + 2 = 10$ people

3rd table: $3 \times 4 + 2 = 14$ people

$\therefore$ number of tables $\times 4 + 2$ is the correct rule.

2. Number and Algebra, NAP-08600

Let x = number of men players

$$\therefore x + (x - 21) = 123$$

$$2x = 144$$

$$x = 72$$

3. Number and Algebra, NAP-32123

$$\begin{aligned} W &= 300 + 0.05 \times 32000 \\ &= 300 + 1600 \\ &= 1900 \end{aligned}$$

4. Number and Algebra, NAP-02715

$$\begin{aligned} \text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{5 - (-2)}{0 - 3} \\ &= -\frac{7}{3} \end{aligned}$$

5. Number and Algebra, NAP-02691

$$\text{Sun} = \frac{1090}{1200} = 0.91 \text{ ¢/mL}$$

$$\text{Sport} = \frac{540}{600} = 0.9 \text{ ¢/mL}$$

$$\text{Block} = \frac{475}{500} = 0.95 \text{ ¢/mL}$$

$$\text{Wet} = \frac{740}{750} = 0.99 \text{ ¢/mL}$$

$\therefore$ Sport sunscreen is the cheapest per litre.

STRATEGY: Finding the cost per mL is much more efficient in answering this question than finding the cost per litre.

6. Number and Algebra, NAP-08629

$$9 + 4.5x = 90$$

$$4.5x = 81$$

$$\begin{aligned} x &= \frac{81}{4.5} \\ &= 20 \end{aligned}$$

7. Number and Algebra, NAP-55673

$$\text{Total millilitres} = \frac{475}{1.5} = 316.67 \dots$$

$$\begin{aligned} \text{Sand flow} &= \frac{316.67 \dots}{15} \\ &= 21.1 \text{ ml per minute} \end{aligned}$$

8. Number and Algebra, NAP-20326

$$\begin{aligned}\text{Interest earned} &= 530 - 500 \\ &= \$30\end{aligned}$$

$$\begin{aligned}\therefore \text{Interest rate (annual)} \\ &= \frac{30}{500} \\ &= 0.06 \\ &= 6\%\end{aligned}$$